IAT New Test 11

Question paper, answer key and worked solutions

60 questions 180 minutes 240 marks

Biology 1–15; Chemistry 16–30; Mathematics 31–45; Physics 46–60.

Correct +4; incorrect -1; unattempted 0. All 60 questions are available.

This is a provisional 2026-pattern authoring draft for 2027 preparation. Independent academic review is pending.

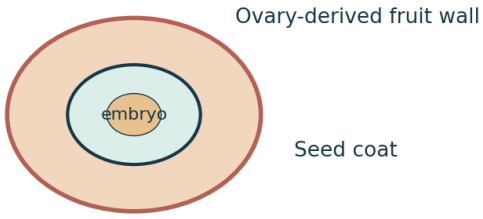
Instructions

Each item has exactly one correct option. Use the separate solution section for review after attempting the paper. **Figures and equations are part of the questions.**

Biology

Q1 The seed-section figure shows an embryo enclosed by a seed coat and a fruit wall derived from the ovary. Which inference is best supported?

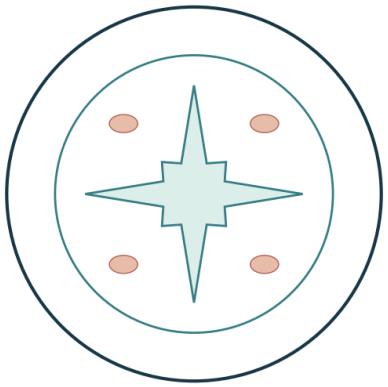
Angiosperm fruit and seed section



A) The plant must be a gymnosperm
B) The ovule was enclosed in an ovary before fruit development
C) The seed contains no embryo
D) The fruit wall formed from the pollen grain

Q2 Use the root cross-section in the figure. Which identification and vascular arrangement fit?

Young root transverse section



Central xylem star; phloem between arms

A) Monocot root with many radial xylem poles
B) Dicot stem with scattered bundles
C) Dicot root with radial xylem and phloem
D) Monocot stem with closed collateral bundles

Q3 Two glucose residues join by a covalent bond after removal of one H_2O molecule. Which linkage type is formed in a disaccharide?
A) Peptide bond
B) Phosphodiester bond
C) Ester bond
D) Glycosidic bond

Q4 Two leaves at the same light and temperature have equal stomatal aperture. Leaf X is in dry air; Y is in humid air. X loses more water while short-term CO_2 uptake is similar. Which mechanism best explains the divergence?
A) Steeper water-vapour gradient drives greater transpiration from X
B) Dry air makes the stomatal pore physically wider despite the stated equality
C) Humid air abolishes photosynthesis entirely
D) Water moves from air into X because its vapour pressure is lower

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Q5 At an active tissue, CO_2 rises and pH falls while the incoming arterial blood is unchanged. What effect on haemoglobin helps deliver more O_2 locally?
A) A left shift and stronger O_2 binding
B) A right shift and lower O_2 affinity
C) No change because blood pH cannot vary
D) Conversion of O_2 into CO_2 inside haemoglobin

Q6 A self-pollinated flower receives viable pollen that germinates on its stigma but its tube stops in the style; unrelated pollen reaches the ovule and seeds form. What mechanism best fits the control?
A) Failure to produce pollen
B) Absence of any functional ovule
C) Self-incompatibility rejecting self pollen after germination
D) Mechanical absence of a stigma

Q7 Two equally large isolated populations each follow Hardy–Weinberg proportions: their A-allele frequencies are 0.8 and 0.2. They are pooled for sampling before any cross-population mating. How does the pooled heterozygote fraction compare with the Hardy–Weinberg expectation calculated from pooled p=0.5?
A) Observed 0.50; expected 0.50
B) Observed 0.68; expected 0.50
C) Observed 0.25; expected 0.32
D) Observed 0.32; expected 0.50

Q8 In the fermentation figure, identify microbes P and Q from their observed products. Which pair is consistent?

Two food fermentations

P: dough → ethanol + CO_2

Q: milk → lactic acid in curd

A) P is yeast; Q is Lactobacillus
B) P is Lactobacillus; Q is yeast
C) Both are methanogenic archaea
D) P is a virus; Q is a cyanobacterium

Q9 The figure shows a vector opened with two different restriction enzymes, E and H, and an insert prepared with matching E and H ends. Their sticky overhang sequences are noncompatible with one another. Why is insertion favoured in one orientation?

Two noncompatible sticky ends

Vector: E / 5′–AATT ... H / 5′–AGCT

Insert: E-compatible left ... H-compatible right

E and H overhangs do not pair with each other

A) Both ends are identical, so either orientation works
B) Each insert end can anneal only to its corresponding vector end, then ligase seals the aligned backbone
C) The E end can ligate to both E and H sites equally
D) The insert has no phosphates, so orientation is irrelevant

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Q10 A population grows by 20% annually at low density, but in later years additions diminish as size nears a stable level while births continue. Which description fits the later regime?
A) Constant exponential growth with unlimited resources
B) Immediate population extinction
C) Density-limited approach to carrying capacity
D) A fixed annual decrease regardless of density

Q11 A pesticide-resistance allele is rare before spraying. After repeated exposure, resistant insects survive and leave more offspring, and the allele frequency rises. Which process is directly illustrated?
A) Directed mutation induced in every insect by need
B) Random drift without differential survival
C) Acquired resistance inherited because individual insects practised tolerance
D) Natural selection acting on heritable variation

Q12 Two independent genes A/a and B/b affect a visible trait. An AaBb individual is test-crossed with aabb. If segregation and assortment are independent, what fraction of offspring are AaBb?
A) 1/4
B) 1/2
C) 3/4
D) 1

Q13 A substance is filtered into a nephron at 100 mg min^{-1}, is neither secreted nor metabolised, and appears in urine at 25 mg min^{-1}. How much is reabsorbed per minute?
A) 25 mg
B) 75 mg
C) 100 mg
D) 125 mg

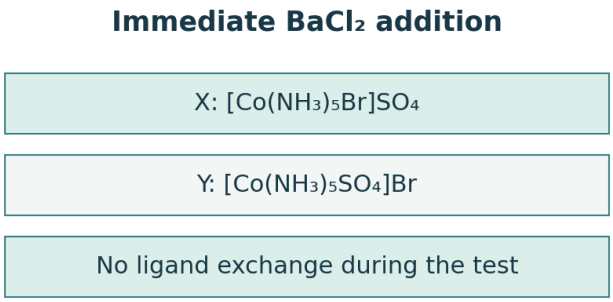
Q14 Membrane vesicles have more glucose inside than outside. At equal pH and zero membrane potential, glucose uptake occurs only when Na^+ concentration is higher outside than inside; no ATP is supplied. Collapsing that sodium concentration difference stops uptake. Which mechanism best explains the observations?
A) Glucose diffuses down its own concentration gradient
B) A glucose ATPase directly hydrolyses supplied ATP
C) Secondary active co-transport couples inward Na^+ movement to uphill glucose uptake
D) Glucose is synthesised inside rather than transported

Q15 In a young dicot stem, a cambial ring forms from vascular-cambium strips within bundles and tissue between bundles. What is the origin of that tissue between bundles?
A) Dead vessel elements in xylem
B) Epidermal cuticle
C) Leaf mesophyll
D) Interfascicular parenchyma that becomes meristematic

Chemistry

Q16 Which proposed resonance contributor for the carbonate ion CO_3^{2-} is valid under the usual octet Lewis description?
A) One C=O and two C–O^- bonds, carbon with eight electrons
B) Three C=O double bonds at one carbon
C) One C–O bond and two unbonded oxygen atoms
D) A carbonate contributor with a carbon radical

Q17 The figure shows ionisation isomers X and Y. Their aqueous samples receive $BaCl_2$ separately. Which sample yields immediate $BaSO_4$ from a free counter-ion?



A) Only Y
B) Only X
C) Both X and Y
D) Neither

Q18 Across the lanthanoid series, the ionic radii of common Ln^{3+} ions generally decrease. What causes this trend?
A) Progressive filling of 6p orbitals
B) Falling nuclear charge
C) Weak shielding by added 4f electrons as nuclear charge increases
D) Elimination of all inner-shell electrons

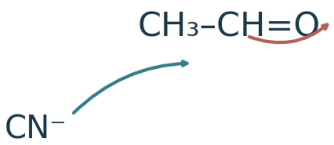
Q19 Among O^{2-}, F^-, Na^+ and Mg^{2+}, all with ten electrons, which ion has the smallest radius?
A) O^{2-}
B) F^-
C) Na^+
D) Mg^{2+}

Q20 For hydrogen, what is the limiting photon energy of the Balmer series as the upper level n approaches infinity? Use $E_n = -13.6/n^2$ eV.
A) 3.4 eV
B) 13.6 eV
C) 1.89 eV
D) 10.2 eV

Q21 A 250 mL sample of 0.40 M acid HA is diluted to 500 mL. A 100 mL aliquot of the diluted acid reacts 1:1 with NaOH. How many moles of NaOH are needed for complete neutralisation?
A) 0.010 mol
B) 0.020 mol
C) 0.040 mol
D) 0.100 mol

Q22 Follow the electron-flow arrows in the carbonyl addition figure. Which role does CN^- play in the bond-forming step?

Electron-pair movement in ethanal addition



CN pair attacks carbonyl carbon; π electrons move to O

A) Electrophile accepting an electron pair
B) Leaving group departing from a saturated carbon
C) Nucleophile donating a pair to an electrophilic carbon
D) Brønsted acid donating H^+

Q23 A ring has CH_3 at carbon 1 and Br at carbon 4. Under mild nitration, both groups direct ortho/para but they favour different available positions. Which site is favoured mainly by the more activating group, assuming steric effects do not reverse the directing effect?
A) Carbon 4 already occupied by Br
B) Carbon 3 or 5, meta to CH_3
C) Carbon 1 already occupied by CH_3
D) Carbon 2 or 6, ortho to CH_3

Q24 Under ordinary SN2 conditions with aqueous hydroxide, why does bromobenzene react much more slowly than bromoethane?
A) Bromobenzene's C–Br bond is to an sp^2 carbon and resists backside substitution
B) Bromobenzene contains no polar bond
C) Bromoethane has an aromatic ring
D) Bromobenzene has a more accessible tetrahedral reaction centre

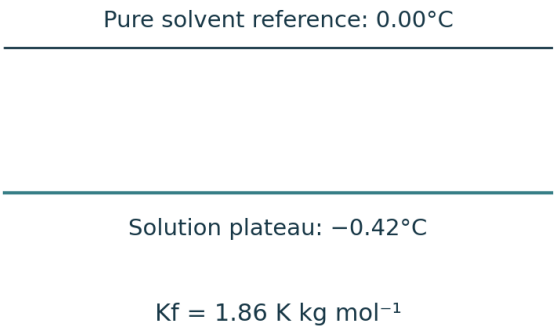
Q25 Anisole (Ph–O–CH_3) is treated with excess concentrated HI under ether-cleavage conditions. Which main pair of products reflects cleavage at the alkyl side?
A) Phenyl iodide and methanol
B) Phenol and iodomethane
C) Benzene and formaldehyde
D) Aniline and methane

Q26 Sucrose solution gives a negative Benedict test before hydrolysis but becomes positive after complete acid hydrolysis followed by neutralisation. What accounts for the change?
A) Sucrose is initially an aldehyde that acid removes
B) Hydrolysis converts both monomers to non-sugars
C) Hydrolysis releases reducing monosaccharides with available anomeric centres
D) Benedict reagent directly hydrolyses sucrose

Q27 A monoprotic weak acid has analytical concentration 0.10 M and measured equilibrium $[H^+]=1.0\times10^{-3}$ M in water, with water autoionisation negligible. What is Ka?
A) 1.0×10^{-3}
B) 1.0×10^{-4}
C) 1.0×10^{-6}
D) 1.01×10^{-5}

Q28 From the freezing-point graph for a 0.10 m electrolyte A_2B solution and the displayed solvent constant, estimate its degree of dissociation into three ions.

Freezing point of 0.10 m A_2B solution



A) About 0.63
B) About 0.26
C) About 0.42
D) About 1.26

Q29 An aqueous $CuSO_4$ cell with inert electrodes is electrolysed so that 0.020 mol electrons pass. Assuming Cu^{2+} plates at the cathode and water oxidises to O_2 at the anode, what are the product amounts?
A) 0.020 mol Cu and 0.020 mol O_2
B) 0.010 mol Cu and 0.005 mol O_2
C) 0.040 mol Cu and 0.010 mol O_2
D) 0.005 mol Cu and 0.010 mol O_2

Q30 Use ΔH_f° (kJ mol^{-1}): $CH_4(g) = -75$, $CO_2(g) = -394$, $H_2O(l) = -286$, $O_2(g) = 0$. What is ΔH° for $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O(l)$?

A) -680

B) $+891$

C) -891

D) -755

Mathematics

Q31 For the ellipse x²/16+y²/12=1, what are its eccentricity and the positive x-directrix coordinate?
A) 1/4 and 16
B) 1/2 and 4
C) √3/2 and 8
D) 1/2 and 8

Q32 A parametric curve has $x = t^2 + 1,\ y = t^3 - 3t$, with $t > 0$. At its horizontal tangent, what are the tangent equation and the value of d^2y/dx^2?
A) $y = -2;\ 3/2$
B) $x = 2;\ 3/2$
C) $y = -2;\ 3$
D) $y = 2;\ -3/2$

Q33 What is the finite area enclosed by y=x² and y=2x−x²?
A) 1/6
B) 1/3
C) 2/3
D) 1

Q34 Consider f: ℝ→[0,∞) defined by f(x)=x². Which classification is correct?
A) Injective but not surjective
B) Both injective and surjective
C) Surjective but not injective
D) Neither injective nor surjective

Q35 For a 3×3 matrix A with det A=−2, what is tr(A adj A)?
A) −2
B) −4
C) 6
D) −6

Q36 Events A and B are independent with P(A)=1/2 and P(B)=1/3. What is P(A∪B)?
A) 2/3
B) 5/6
C) 1/6
D) 1/2

Q37 Five distinct people sit at a round table; rotations count as the same arrangement. How many arrangements have A and B not adjacent?
A) 6
B) 12
C) 18
D) 24

Q38 Write $(1+x)^6 = \sum_{k=0}^{6} c_k x^k$. What is $\sum_{k=0}^{6} k^2 c_k$?
A) 192
B) 480
C) 672
D) 1344

Q39 Let $S_N = \sum_{n=1}^{N} 1/[n(n+1)(n+2)]$. What is the least positive integer N for which S_N>1/5?
A) 1
B) 3
C) 10
D) 2

Q40 For real a, define $L(a) = \lim_{x\to 0}(|x-a|-|a|)/x$, if a finite two-sided limit exists. Which statement is complete?
A) It exists for all a≠0 and equals −sgn(a); at a=0 it fails
B) It exists for all a and equals 0
C) It exists only for a=0 and equals 1
D) It exists for all a>0 and nowhere else

Q41 A quantity y(t)>0 follows dy/dt=2ty and starts at y(0)=3. What is its value at t=1?
A) e^2
B) 3e
C) $3e^2$
D) 6e

Q42 The lines L_1: (x,y,z)=(t,0,0) and L_2: (x,y,z)=(0,1+s,1) extend indefinitely. What is their shortest distance?
A) 0
B) √2
C) 1
D) 2

Q43 For vectors a=(1,1,0), b=(0,1,1), c=(1,0,k), what k makes a,b,c coplanar?
A) 1
B) 0
C) −2
D) −1

Q44 Values 1,2,4 have frequencies 1,2,1. Find the mean absolute deviation from their arithmetic mean.
A) 7/8
B) 3/4
C) 1
D) 9/4

Q45 Using principal ranges arcsin∈[−π/2,π/2], arccos∈[0,π], arctan∈(−π/2,π/2), evaluate arcsin(sin 7π/6)+arccos(cos 7π/6)+arctan(tan 7π/6).
A) π/2
B) 5π/6
C) 7π/6
D) π

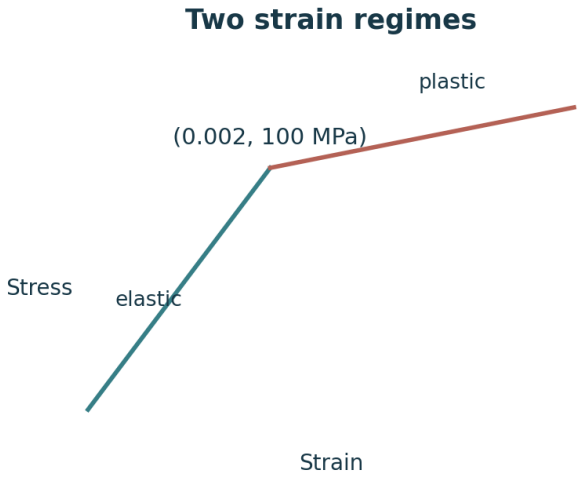
Physics

Q46 A 2 kg block rests on the floor of a lift accelerating upward at 2 m s^{-2}. With g=10 m s^{-2}, what normal force does the floor exert?
A) 16 N
B) 20 N
C) 24 N
D) 28 N

Q47 A 1 kg block starts from rest 2 m above a horizontal rough track. After descending a smooth ramp, it travels 3 m across a rough stretch with μ=0.20, then compresses a spring of k=100 N m^{-1} on the same rough surface. With g=10 m s^{-2}, which equation and approximate maximum compression x are correct?
A) $50x^2=20$, x≈0.63 m
B) $50x^2=14$, x≈0.53 m
C) $50x^2+6x=14$, x≈0.47 m
D) $50x^2+2x=14$, x≈0.51 m

Q48 Inside a uniform-density planet of radius R, gravitational acceleration at distance R/3 from the centre is what fraction of surface g?
A) 1/3
B) 1/9
C) 1
D) 3

Q49 Read the two-regime stress–strain plot. Which elastic modulus follows from the initial straight segment?



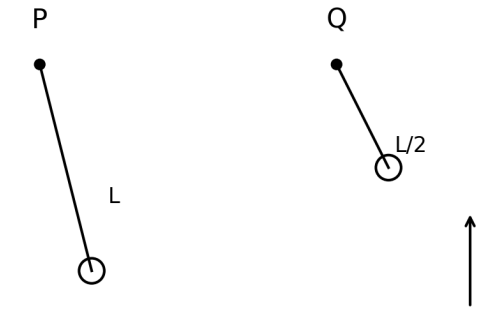
A) 20 GPa
B) 50 GPa
C) 100 GPa
D) 200 GPa

Q50 A 100 g ice sample at 0°C is mixed in an insulated vessel with 100 g water at 50°C. Take water c=4.2 J g^{-1} K^{-1} and ice latent heat L=334 J g^{-1}. What is the final state?
A) All ice melts and water reaches 5°C
B) All ice melts at 0°C
C) About 63 g ice melts; mixture remains at 0°C
D) No ice melts

Q51 An ideal-gas mixture contains equal numbers of molecules of gases X and Y at a common temperature; their molar masses are 4 and 16 units. If vY is the rms speed of species Y, what is the rms speed computed over all molecules in the mixture divided by vY?
A) 1/2
B) 1
C) √2
D) √(5/2)

Q52 Pendulum P has length L in a stationary laboratory. Pendulum Q has length L/2 in a lift accelerating vertically upward at unknown a. Their measured small-oscillation periods satisfy $T_Q/T_P = 1/2$. What are a/g and the apparent weight of a stationary mass m on a scale in the lift?

Infer acceleration from the measured periods



A) 1; 2mg
B) 1/2; 3mg/2
C) 0; mg
D) 3; 4mg

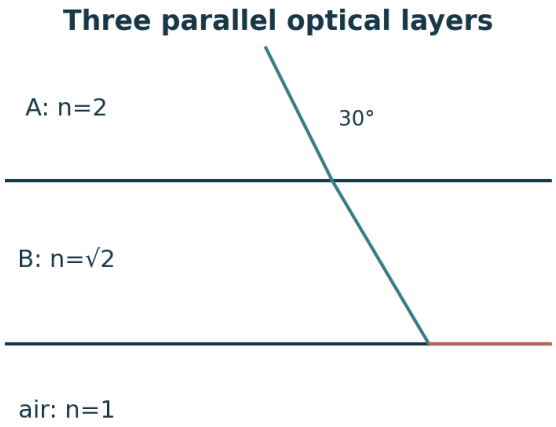
Q53 Capacitors C and 2C are in series across an ideal battery V. While the battery remains connected, dielectric of κ=2 completely fills the C capacitor. What is the ratio of the new series charge to the original series charge?
A) 1
B) 3/2
C) 2/3
D) 2

Q54 A magnetic dipole of magnitude m is in uniform B. Its orientation changes from parallel to perpendicular to B. What change in potential energy occurs?
A) $-mB$
B) 0
C) $+mB$
D) $+2mB$

Q55 A permanent electric dipole p in uniform E is initially perpendicular to the field and rotates into alignment without friction. How do its torque magnitude and potential energy change between these endpoints?
A) Torque increases; energy increases
B) Torque stays maximum; energy decreases
C) Torque falls to zero; energy increases
D) Torque falls to zero; energy decreases

Q56 A Wheatstone bridge has left arms 2 Ω and 3 Ω and right upper arm 4 Ω; the right lower arm is a 5 Ω resistor at 0°C with temperature coefficient 0.004 $°C^{-1}$. At what temperature is the bridge balanced?
A) 50°C
B) 25°C
C) 75°C
D) 100°C

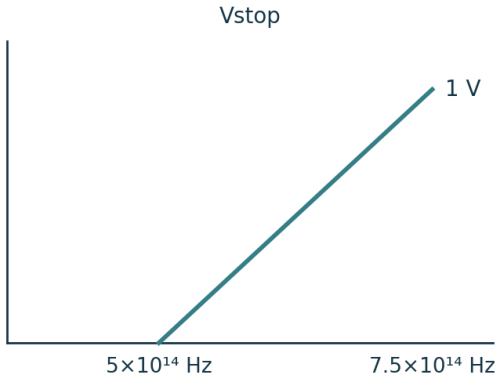
Q57 Using the indices and incoming angle in the parallel-layer diagram, what happens to the ray at the final layer–air interface?



A) Total internal reflection before reaching B
B) It emerges grazing the air surface
C) It emerges into air at 30°
D) It refracts into air at 45°

Q58 From the stopping-potential plot for a metal, infer its work function in electronvolts.

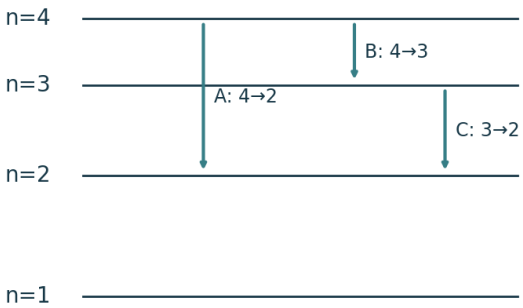
Stopping potential against light frequency



A) 0.5 eV
B) 1 eV
C) 2 eV
D) 4 eV

Q59 One hundred atoms start at hydrogen level n=4. Each reaches n=2 either by direct 4→2 emission A or by the two-step cascade 4→3→2 shown. Exactly 150 photons are emitted in total. Neglect all other paths. What fraction of atoms took the direct path, and what is the mean emitted-photon energy divided by the energy of photon A?

Hydrogen: direct path and cascade



A) 1/2; 1
B) 1/4; 3/4
C) 1/2; 3/2
D) 1/2; 2/3

Q60 A conducting rod of length 1.0 m rotates about one end at 10 rad s^{-1} in a perpendicular uniform magnetic field of 0.20 T. What emf develops between its ends?
A) 1.0 V
B) 2.0 V
C) 0.20 V
D) 10 V

Answer key and worked solutions

Every solution explains the selected choice and the alternative errors.

Biology

Q1 / B angiosperm seed character

Answer: (B) An enclosed ovule.

Why: An angiosperm's ovary encloses ovules and later contributes the fruit wall; an ovule matures into a seed. Seeing both outer fruit wall and internal seed supports this developmental route, not a classification by seed number alone.

Why not the others: (A) gymnosperm seeds lack an enclosing ovary-derived fruit; (C) contradicts the pictured embryo; (D) pollen contributes male gametes rather than fruit wall.

Key point: Trace ovule and ovary into different mature structures.

Q2 / C root vascular arrangement

Answer: (C) Dicot root.

Why: Radial separation of xylem and phloem plus star-shaped central xylem and small pith is characteristic of a young dicot root. It is anatomical evidence about vascular arrangement, not a statement that water and food move in the same cells.

Why not the others: (A) monocot roots usually have many poles and larger pith; (B) stems arrange bundles differently; (D) scattered closed bundles describe a monocot stem.

Key point: Combine stele arrangement and pith before assigning organ type.

Q3 / D biomolecule linkage

Answer: (D) Glycosidic bond.

Why: Condensation between hydroxyl groups on monosaccharides joins them through an oxygen bridge, a glycosidic linkage, while releasing water. The precise anomeric configuration requires structure information beyond this description.

Why not the others: (A) joins amino-acid residues; (B) links nucleotides in a sugar-phosphate backbone; (C) is a different carboxylic-acid/alcohol linkage.

Key point: Identify the constituent monomers before naming their bond.

Q4 / A stomatal exchange

Answer: (A) Steeper vapour gradient.

Why: Equal apertures control stomatal conductance approximately, but dry outside air lowers external water-vapour partial pressure, steepening the outward diffusion gradient. CO_2 uptake may remain similar if internal carbon supply and other limitations compensate; a short observation does not prove identical total photosynthesis.

Why not the others: (B) violates the controlled aperture observation; (C) is unsupported by measured uptake; (D) reverses the stated net water loss direction.

Key point: Separate conductance from the driving partial-pressure difference.

Q5 / B gas transport and dissociation

Answer: (B) Right shift.

Why: Higher CO_2 and H^+ favour haemoglobin conformations with lower oxygen affinity (Bohr effect). At the same tissue oxygen partial pressure, a right-shifted dissociation curve allows more oxygen unloading.

Why not the others: (A) would hinder unloading; (C) ignores local acidification; (D) confuses gas transport and chemical conversion.

Key point: Compare affinity at fixed local oxygen pressure.

Q6 / C pollen compatibility and outbreeding

Answer: (C) Self-incompatibility.

Why: The viable self pollen starts germination but fails later, whereas unrelated pollen travels through the same style and produces seeds. This bypass comparison localises rejection to recognition or growth after contact, rather than a general defect in the female tract.

Why not the others: (A) self pollen was present and viable; (B) cross-pollination produced seeds; (D) pollen germinated on the stigma.

Key point: A compatible cross controls for general floral viability.

Q7 / D Hardy-Weinberg assumption

Answer: (D) Observed 0.32 versus expected 0.50.

Given: Each population has $2pq=2(0.8)(0.2)=0.32$ heterozygotes; equal weighting preserves 0.32 in the pooled sample.

Formula/Principle: The pooled allele frequency is 0.5, whose random-mating expectation would be $2(0.5)(0.5)=0.50$.

Step 1: Population structure without interbreeding causes a heterozygote deficit relative to a single randomly mating population.

Why: Hardy–Weinberg genotype frequencies require random mating within the population considered. Combining two internally random-mating groups with different allele frequencies before cross-mating does not create the heterozygotes predicted by pooled allele frequency. This is a structure effect, not evidence that an allele changed.

Why not the others: (A) assumes pooled random mating already occurred; (B) overcounts heterozygotes; (C) confuses an allele frequency or homozygote share with heterozygote fraction.

Key point: Check the population boundary before applying 2pq.

Q8 / A microbe product identification

Answer: (A) Yeast and Lactobacillus.

Why: Yeast fermentation produces CO_2 that leavens dough and ethanol, while lactic-acid bacteria convert milk sugars to lactic acid and lower curd pH. The paired product evidence distinguishes both organisms better than a single habitat clue.

Why not the others: (B) reverses the characteristic fermentations; (C) methanogens produce methane under different conditions; (D) neither proposed organism matches both food processes.

Key point: Match microbes to products and substrate context.

Q9 / B restriction recognition and ends

Answer: (B) Distinct compatible end pairs enforce orientation.

Why: The E overhang of insert pairs with vector E, and H pairs with H; swapping the insert would place noncomplementary E opposite H at both junctions. Complementary annealing establishes the allowed geometry, then ligase closes phosphodiester bonds. This reduces reverse insertion and simple self-ligation relative to using one identical site, although experimental verification remains necessary.

Why not the others: (A) contradicts the nonidentical ends; (C) ignores sequence-specific pairing; (D) a phosphate backbone is required for covalent sealing and does not remove orientation constraints.

Key point: Two different noncompatible overhangs give directional cloning.

Q10 / C population growth regime

Answer: (C) Density limitation.

Why: Slowing increments near a stable size imply per-capita net growth declines as density increases and limiting resources or interactions become important. Births can continue while balanced by mortality; a plateau does not mean reproduction stops.

Why not the others: (A) would retain proportional growth; (B) contradicts stable persistence; (D) predicts falling numbers rather than diminishing increments.

Key point: Compare per-capita growth across density levels.

Q11 / D evolutionary selection evidence

Answer: (D) Natural selection.

Why: A pre-existing heritable variant becomes common when the environment changes relative survival and reproduction. The frequency trend and survival comparison support selection; they do not show that pesticide intentionally created an allele.

Why not the others: (A) asserts need-directed mutation; (B) ignores differential fitness; (C) invokes inheritance of acquired training.

Key point: Selection changes variant frequency through unequal reproductive success.

Q12 / A Mendelian two-locus segregation

Answer: (A) One quarter.

Given: AaBb produces AB,Ab,aB,ab gametes equally; aabb produces ab only.

Formula/Principle: P(AB gamete)=P(A)P(B)=(1/2)(1/2).

Step 1: AB×ab produces AaBb in 1/4 of offspring.

Why: The recessive tester exposes each gamete from the heterozygote directly in offspring genotype. Independent assortment is the reason the two halves multiply; linkage would alter this prediction.

Why not the others: (B) counts only one locus; (C) counts any heterozygosity; (D) treats every gamete as AB.

Key point: Decode a two-locus test cross by gamete types.

Q13 / B nephron clearance and reabsorption

Answer: (B) 75 mg min^{-1}.

Given: Filtered 100 mg min^{-1}; excreted 25 mg min^{-1}; secretion zero.

Formula/Principle: Excretion=filtration−reabsorption+secretion.

Step 1: Reabsorption=100−25=75 mg min^{-1}.

Why: No secretion or metabolism occurs. Thus, the missing filtered material must return to blood, so the reabsorbed amount is determined. The 25 mg/min appearing in urine is the residual after this return from the filtered load.

Why not the others: (A) repeats excretion; (C) assumes no urine loss; (D) adds rather than subtracts.

Key point: Use a mass balance with all transport routes stated.

Q14 / C Direction and energy source of gradient-coupled uptake

Answer: (C) Gradient-coupled secondary active co-transport.

Why: At zero voltage, the higher external Na^+ concentration favours inward sodium movement, whereas the stated glucose concentrations oppose passive inward glucose movement. Coupling those movements can use the sodium gradient's free energy without direct ATP hydrolysis by the transporter. Losing the imposed gradient removes the energy source and explains why uptake stops.

Why not the others: (A) predicts the opposite direction for glucose's own gradient; (B) requires supplied ATP, which is absent; (D) does not explain observed uptake from the medium or dependence on the sodium gradient.

Key point: State both gradient direction and voltage before inferring coupled transport.

Q15 / D stem cambium origin

Answer: (D) Interfascicular parenchyma.

Why: Parenchyma between vascular bundles can regain meristematic activity, linking the intrafascicular cambial segments into a continuous ring. That ring produces secondary vascular tissues; the question asks its origin, not the final products.

Why not the others: (A) differentiated dead xylem cannot divide; (B) is an external protective layer; (C) is leaf photosynthetic tissue.

Key point: Distinguish cambium formed inside and between stem bundles.

Chemistry

Q16 / A resonance contributor validity

Answer: (A) One double and two single bonds.

Why: Carbon's octet accommodates one double bond and two single bonds to oxygen. Two singly bonded oxygens each carry formal charge −1; alternative equivalent contributors put the double bond at another oxygen without moving nuclei.

Why not the others: (B) puts twelve bonding electrons around second-period C; (C) loses the three connected C–O bonds; (D) invents an unpaired electron absent from the ion.

Key point: Resonance moves electrons, not atoms, and obeys octets.

Q17 / B ionisation isomer wet test

Answer: (B) Only X.

Why: Sulfate lies outside the coordination sphere of X as an exchangeable counter-ion, so Ba^{2+} precipitates it. In Y sulfate is coordinated and bromide is the free counter-ion; the immediate sulfate test distinguishes ionisation isomers without requiring dissociation of the complex.

Why not the others: (A) reverses inner and outer sphere; (C) treats coordinated sulfate as free; (D) ignores X's counter-ion.

Key point: Locate the ion outside square brackets before predicting a rapid wet test.

Q18 / C lanthanoid contraction trend

Answer: (C) Weak 4f shielding.

Why: Additional protons attract the electron cloud more strongly, while added 4f electrons incompletely screen that charge from outer electrons. The resulting effective nuclear-charge rise contracts successive trivalent ions.

Why not the others: (A) misidentifies the subshell filled; (B) nuclear charge increases; (D) inner shells remain occupied.

Key point: Compare increasing nuclear charge with incomplete 4f screening.

Q19 / D isoelectronic-radius constraint

Answer: (D) Mg^{2+}.

Why: Isoelectronic ions have the same electron count but different proton counts. Mg^{2+} has the largest nuclear charge among the four, so it attracts the common ten-electron cloud most strongly, producing the smallest radius.

Why not the others: (A) has only eight protons; (B) nine; (C) eleven versus magnesium's twelve.

Key point: Within an isoelectronic series, radius decreases with nuclear charge.

Q20 / A spectral series limit

Answer: (A) 3.4 eV.

Given: Balmer transitions terminate at n=2.

Formula/Principle: Photon energy=$13.6(1/2^2-1/n^2)$ eV.

Step 1: As $n\to\infty$, second term tends to zero, giving 13.6/4=3.4 eV.

Worked derivation:

$$\Delta E_{\infty\to 2} = 13.6(1/2^2 - 0) = 3.40 \text{ eV}.$$

Why: The series limit corresponds to excitation from n=2 to the continuum, not from the ground state. The photon energy approaches a finite maximum even though the upper quantum number is unbounded.

Why not the others: (B) is the ground-state ionisation energy; (C) is a finite Balmer gap; (D) is the first Lyman gap.

Key point: Identify the series' fixed lower level first.

Q21 / B solution dilution and reaction yield

Answer: (B) 0.020 mol.

Given: Initial acid 0.250×0.40=0.100 mol; dilution gives 0.20 M.

Formula/Principle: Aliquot acid moles=0.100 L×0.20 M; neutralisation ratio 1:1.

Step 1: 0.020 mol HA in aliquot, hence 0.020 mol NaOH.

Worked derivation:

$$n_{\text{stock}} = 0.250(0.40) = 0.100 \text{ mol},$$
$$C_{\text{diluted}} = 0.100/0.500 = 0.200 \text{ M},$$
$$n_{\text{NaOH}} = n_{\text{aliquot}} = 0.100(0.200) = 0.0200 \text{ mol}.$$

Why: Dilution leaves the whole batch's acid moles unchanged but spreads them over twice the volume. Only one fifth of the final batch is titrated, so using all 0.100 mol would overcount.

Why not the others: (A) halves the aliquot amount again; (C) uses undiluted concentration; (D) treats the entire batch as the aliquot.

Key point: Recalculate concentration, then use aliquot volume.

Q22 / C electrophile nucleophile selectivity

Answer: (C) Nucleophile.

Why: The carbonyl bond is polarised toward oxygen, leaving carbon electron-deficient. Cyanide supplies an electron pair to form a new C–C bond while π electrons move to oxygen, the defining electron-flow pattern for nucleophilic addition.

Why not the others: (A) reverses electron-pair flow; (B) no departure is shown; (D) CN^- does not donate a proton here.

Key point: Follow curved-arrow electron donation, not reagent charge alone.

Q23 / D aromatic substitution orientation

Answer: (D) Carbon 2 or 6.

Why: Methyl is activating and ortho/para directing; at a para-bromotoluene framework its para site is occupied, so the equivalent positions 2 and 6 are available. Bromine also directs ortho/para but deactivates the ring, making its preference at positions 3/5 less influential under the stated comparison.

Why not the others: (A) and (C) are already substituted sites; (B) follows bromine's direction while ignoring stronger methyl activation.

Key point: Resolve competing directing effects using relative activation and occupancy.

Q24 / A aryl versus alkyl reactivity

Answer: (A) Aryl sp^2 carbon resists SN2.

Why: SN2 requires backside approach to a tetrahedral carbon; the aromatic sp^2 carbon and C–Br bond with partial double-bond character obstruct that path. The ordinary primary alkyl bromide offers an accessible saturated centre for displacement.

Why not the others: (B) the aryl C–Br bond is polar; (C) bromoethane is aliphatic; (D) reverses the structural comparison.

Key point: Match the reaction mechanism to the carbon bearing the leaving group.

Q25 / B ether cleavage

Answer: (B) Phenol and CH_3I.

Why: Protonation of the ether oxygen makes cleavage easier; iodide attacks the methyl carbon through an accessible SN2 path. The aryl C–O bond is not cleaved by ordinary backside attack on sp^2 carbon, leaving phenol.

Why not the others: (A) requires aryl substitution and the wrong alcohol; (C) lacks the iodide product; (D) changes oxygen to nitrogen without a source.

Key point: Unsymmetrical aryl–alkyl ethers cleave at the alkyl carbon.

Q26 / C reducing sugar before-after hydrolysis

Answer: (C) Reducing products after hydrolysis.

Why: Sucrose links both anomeric carbons, preventing the free hemiacetal/hemiketal forms needed for reduction in the initial assay. Hydrolysis liberates glucose and fructose, which can participate in reducing-sugar chemistry under the alkaline test conditions.

Why not the others: (A) sucrose is initially nonreducing; (B) products remain sugars; (D) the stipulated acid hydrolysis occurs before the neutralised test.

Key point: Check whether anomeric carbons are tied up in the glycosidic bond.

Q27 / D weak-acid dissociation

Answer: (D) About 1.01×10^{-5}.

Given: $[A^-]=x=0.001$ M, $[HA]=0.10-x=0.099$ M.

Formula/Principle: $K_a=x^2/(C-x)$.

Step 1: $10^{-6}/0.099\approx1.0101\times10^{-5}$.

Worked derivation:

$$K_a=\frac{x^2}{C-x}=\frac{(10^{-3})^2}{0.10-0.001}=\frac{10^{-6}}{0.099}\approx1.01\times10^{-5}.$$

Why: The measured hydrogen concentration equals the dissociation extent for the specified monoprotic acid. Keeping the small depletion in the denominator provides the stated precise value rather than just an order-of-magnitude approximation.

Why not the others: (A) reports x; (B) moves the decimal place one too far; (C) omits the denominator.

Key point: Use equilibrium concentrations rather than analytical concentration blindly.

Q28 / A van Hoff factor inference

Answer: (A) About 0.63.

Given: $\Delta T_f=0.42$ K, $K_f=1.86$, $m=0.10$; complete dissociation would produce three ions.

Formula/Principle: $i=\Delta T_f/(K_f m)=1+2\alpha$ for $A_2B\rightarrow2A+B$.

Step 1: $i=0.42/0.186\approx2.258$; $\alpha=(i-1)/2\approx0.629$.

Worked derivation:

$$i=\frac{0.42}{1.86(0.10)}\approx2.258,\quad \alpha=\frac{i-1}{3-1}\approx0.629.$$

Why: The observed depression exceeds the non-electrolyte value but is below the three-particle full-dissociation limit. Each dissociated formula unit adds two net particles relative to one undissociated unit.

Why not the others: (B) treats $i-1$ as a different ion count; (C) copies ΔT; (D) takes $i-1$ without dividing by two and exceeds unity.

Key point: Infer particle factor before converting it into dissociation fraction.

Q29 / B electrolysis discharge order

Answer: (B) 0.010 mol Cu, 0.005 mol O_2.

Given: 0.020 mol electrons; $Cu^{2+}+2e^- \rightarrow Cu$, and water oxidation releases one O_2 per four electrons.

Formula/Principle: Product moles=electron moles divided by electrons per product.

Step 1: Cu=0.020/2=0.010 mol; O_2=0.020/4=0.005 mol.

Worked derivation:

$$n_{\mathrm{Cu}} = 0.020/2 = 0.010 \text{ mol}, \quad n_{\mathrm{O_2}} = 0.020/4 = 0.005 \text{ mol}.$$

Why: Cathode and anode consume or release the same charge but have distinct electron stoichiometries. Assuming inert electrodes avoids metal-anode dissolution, which would change the products.

Why not the others: (A) uses one electron per product; (C) multiplies instead of divides for Cu; (D) exchanges the two denominators.

Key point: Balance both electrode half reactions separately.

Q30 / C Hess-cycle route

Answer: (C) −891 kJ mol^{-1}.

Given: Formation enthalpies for products and reactants.

Formula/Principle: $\Delta H_{rxn} = \Sigma \nu \Delta H_f(\text{products}) - \Sigma \nu \Delta H_f(\text{reactants})$.

Step 1: [−394+2(−286)]−[−75+0]=−966+75=−891.

Worked derivation:

$$\Delta H^\circ = [-394 + 2(-286)] - [-75 + 2(0)] = -891 \text{ kJ mol}^{-1}.$$

Why: Hess's law combines state-function formation enthalpies with balanced stoichiometric coefficients. Including the physical state of water matters because liquid and vapour have different formation values.

Why not the others: (A) omits water or methane contributions; (B) reverses sign; (D) uses an incorrect product sum.

Key point: Multiply each formation value by its reaction coefficient.

Mathematics

Q31 / D ellipse focus-directrix

Answer: (D) 1/2 and x=8.

Given: $a^2=16$, $b^2=12$ for major axis x.

Formula/Principle: $c^2=a^2-b^2$; $e=c/a$; directrix $x=a/e$.

Step 1: c=2, a=4, e=1/2, x=8.

Worked derivation:

$$c = \sqrt{16-12} = 2, \quad e = \frac{2}{4} = \frac{1}{2}, \quad x_{\text{directrix}} = \frac{a}{e} = 8.$$

Why: The focus distance is not itself eccentricity; it must be normalised by the semimajor axis. The positive directrix lies outside the right vertex x=4.

Why not the others: (A) divides twice; (B) uses the vertex as directrix; (C) confuses b/a with eccentricity.

Key point: Compute c, then e, then the directrix.

Q32 / A Parametric horizontal tangent and second derivative

Answer: (A) $y = -2$, with second derivative $3/2$.

Given: $dx/dt = 2t > 0$, so differentiation with respect to x is valid.

Formula/Principle:

$$\frac{dy}{dx} = \frac{3t^2 - 3}{2t}.$$

Step 1: A horizontal tangent requires $t^2 = 1$; the domain selects $t = 1$, giving $(x, y) = (2, -2)$.

Step 2:

$$\frac{d}{dt}\left(\frac{dy}{dx}\right) = \frac{3}{2}(1 + t^{-2}).$$

Step 3:

$$\frac{d^2y}{dx^2} = \frac{3(1 + t^{-2})}{4t}, \qquad \left.\frac{d^2y}{dx^2}\right|_{t=1} = \frac{3}{2}.$$

Worked derivation:

$$\text{tangent: } y = -2, \qquad y'' = \frac{3}{2} > 0.$$

Why: The second parametric derivative must be divided by dx/dt once more; it is not merely the derivative of the slope with respect to t. The positive parameter restriction removes the other algebraic stationary parameter.

Why not the others: (B) gives a vertical line; (C) omits the second division by $dx/dt = 2$; (D) uses the excluded negative parameter.

Key point: Convert each derivative from parameter to coordinate.

Q33 / B area between curves

Answer: (B) 1/3.

Given: Intersections satisfy x²=2x−x².

Formula/Principle: Area=∫(upper−lower) dx between crossings.

Step 1: 2x(x−1)=0 gives x=0,1; upper is 2x−x² between them.

Worked derivation:

$$A = \int_0^1 [(2x - x^2) - x^2]dx = \int_0^1 (2x - 2x^2)dx = \left[x^2 - \frac{2}{3}x^3\right]_0^1 = \frac{1}{3}.$$

Why: Both curves share the crossing endpoints, and the linear-minus-quadratic curve is above in between. Integrating only one curve or reversing their order changes the enclosed area.

Why not the others: (A) halves the difference; (C) doubles the area; (D) uses a bounding rectangle.

Key point: Solve intersections and decide vertical order before integration.

Q34 / C injection versus surjection

Answer: (C) Onto, not one-to-one.

Worked derivation:

$$f(a) = f(-a)\ (a \neq 0), \qquad \forall y \geq 0,\ f(\sqrt{y}) = y.$$

Why: Opposite nonzero inputs collide, so injection fails. Every nonnegative codomain value has a real preimage, including zero, so the function is surjective onto the expressly stated codomain.

Why not the others: (A) reverses both properties; (B) overlooks ±a; (D) treats the codomain as all reals.

Key point: Onto depends on the specified codomain.

Q35 / D adjoint identity

Answer: (D) −6.

Given: A is 3×3 and detA=−2.

Formula/Principle: A adj A=(det A)I₃.

Step 1: Trace of −2I₃ is three times −2.

Worked derivation:

$$\mathrm{tr}(A\,\mathrm{adj}\,A) = \mathrm{tr}[-2I_3] = -2 \cdot 3 = -6.$$

Why: The adjugate identity converts the whole matrix product to a scalar multiple of the identity; no individual entries of A are needed. The dimension enters through trace, not determinant.

Why not the others: (A) reports the scalar alone; (B) assumes two dimensions; (C) drops the negative sign.

Key point: Apply the adjugate identity before taking trace.

Q36 / A independent versus exclusive events

Answer: (A) 2/3.

Given: Independent events can overlap.

Formula/Principle: P(A∩B)=P(A)P(B); union subtracts intersection once.

Step 1: Intersection=1/6; union=1/2+1/3−1/6=2/3.

Worked derivation:

$$P(A \cup B) = \frac{1}{2} + \frac{1}{3} - \left(\frac{1}{2}\right)\left(\frac{1}{3}\right) = \frac{2}{3}.$$

Why: Independence supplies a nonzero overlap, unlike mutual exclusivity. Adding probabilities without subtracting overlap counts outcomes in both events twice. Their intersection is exactly the product of the two nonzero event probabilities under the given independence condition.

Why not the others: (B) omits overlap correction; (C) gives intersection only; (D) reports P(A).

Key point: Independent and mutually exclusive are different relations.

Q37 / B circular arrangement with forbidden pair

Answer: (B) 12.

Given: Rotational symmetry but reflections remain distinct.

Formula/Principle: Total (5−1)!; adjacent pair as a block 2!(4−1)!.

Step 1: 24−12=12.

Worked derivation:

$$N = (5-1)! - 2(4-1)! = 24 - 12 = 12.$$

Why: Fixing one person's position removes rotation. Counting an AB block in both orders captures every forbidden adjacency exactly once for five distinct people.

Why not the others: (A) halves for reflections incorrectly; (C) subtracts too few forbidden cases; (D) includes adjacent pairs.

Key point: Use a block for circular adjacency, then complement.

Q38 / C Weighted binomial coefficient sums via finite differentiation

Answer: (C) 672.

Given: $c_k = \binom{6}{k}$ are coefficients of a finite polynomial.

Formula/Principle: Differentiate the polynomial once and twice, then use $k^2 = k(k-1) + k$.

Step 1:

$$\sum kc_k = 6(1+1)^5 = 192.$$

Step 2:

$$\sum k(k-1)c_k = 30(1+1)^4 = 480.$$

Step 3:

$$\sum k^2 c_k = 480 + 192 = 672.$$

Worked derivation:

$$\sum_{k=0}^{6} k^2 \binom{6}{k} = 6 \cdot 5 \cdot 2^4 + 6 \cdot 2^5 = 672.$$

Why: Ordinary coefficient sums lose the weighting by k; differentiation creates that weighting exactly. The second derivative generates $k(k-1)$, so the first-derivative contribution must be restored to obtain k^2.

Why not the others: (A) uses only the first derivative; (B) uses only the second; (D) doubles the complete sum unnecessarily.

Key point: Build polynomial weights from derivatives of the binomial identity.

Q39 / D telescoping rational sum

Answer: (D) N=2.

Given: Three consecutive factors and a strict threshold for the partial sum.

Formula/Principle: 1/[n(n+1)(n+2)]=(1/2)[1/(n(n+1))−1/((n+1)(n+2))].

Step 1: S_N=1/4−1/[2(N+1)(N+2)].

Step 2: S_N>1/5 requires (N+1)(N+2)>10, first true at N=2.

Worked derivation:

$$\frac{1}{4} - \frac{1}{2(N+1)(N+2)} > \frac{1}{5} \iff (N+1)(N+2) > 10; \quad N_{\min} = 2.$$

Why: The rational blocks telescope to a formula retaining a positive endpoint correction. The strict inequality rejects N=1 because (2)(3)=6, while N=2 has (3)(4)=12; the question asks the first crossing, not the sum at a fixed ten terms.

Why not the others: (A) gives S_1=1/6 below 1/5; (B) works but is not least; (C) also works but is not least.

Key point: Derive the finite remainder, then solve the threshold inequality.

Q40 / A one-sided absolute-value limit

Answer: (A) The limit exists exactly for $a \neq 0$, and equals $-\operatorname{sgn}(a)$.

Given: The absolute value is locally smooth unless its argument crosses zero at the limiting point.

Formula/Principle: Treat $a > 0,\ a < 0,\ a = 0$ separately.

Step 1: For $a > 0$, sufficiently small x gives $|x - a| = a - x$, so

$$L(a) = \lim_{x \to 0} (-x)/x = -1.$$

Step 2: For $a < 0$, sufficiently small x gives $|x - a| = x - a$ and $|a| = -a$, so $L(a) = 1$.

Step 3: At $a = 0$,

$$\lim_{x \to 0^+} |x|/x = 1, \quad \lim_{x \to 0^-} |x|/x = -1,$$

so the two-sided limit fails.

Worked derivation:

$$L(a) = \begin{cases} -1, & a > 0, \\ 1, & a < 0, \end{cases} \quad L(0) \text{ does not exist.}$$

Why: The sign of a determines which linear branch applies near zero. Testing a=0 separately is necessary because its cusp remains at the limiting point rather than outside the local neighbourhood.

Why not the others: (B) ignores both branch slopes; (C) mistakes a one-sided value at the cusp for a two-sided limit; (D) omits the valid negative-parameter branch.

Key point: Separate a moving cusp from the smooth local branches.

Q41 / B separable initial-value problem

Answer: (B) 3e.

Given: Separable first-order IVP.

Formula/Principle: dy/y=2t dt, then use the initial value.

Step 1: ln y=t²+C; y(0)=3 fixes e^C=3.

Worked derivation:

$$y(t) = 3e^{t^2}, \qquad y(1) = 3e.$$

Why: Integrating 2t produces t², not 2t²; the multiplicative constant survives the exponential and is fixed by y(0). The selected expression follows only after both the integration and initial condition.

Why not the others: (A) omits initial factor; (C) uses an incorrect exponent; (D) treats the exponential as a linear factor.

Key point: Apply the initial condition after integration.

Q42 / C skew-line distance

Answer: (C) 1.

Given: L_1 lies in z=0 and L_2 in z=1; their xy projections cross at (0,0).

Formula/Principle: Any joining vector has vertical component 1, hence length ≥1; attain equality when xy coordinates coincide.

Step 1: Choose t=0, s=−1, giving points (0,0,0) and (0,0,1).

Worked derivation:

$$d_{\min} = \|(0,0,1) - (0,0,0)\| = 1.$$

Why: The two lines are skew but their projections meet, so a vertical common perpendicular achieves the lower bound imposed by z separation. An arbitrary pair of points gives a longer segment.

Why not the others: (A) requires intersecting z levels; (B) includes an avoidable horizontal offset; (D) doubles the vertical gap.

Key point: Prove both a distance lower bound and an attaining pair.

Q43 / D scalar triple product

Answer: (D) −1.

Given: Coplanarity means scalar triple product zero.

Formula/Principle: (a×b)·c=0.

Step 1: a×b=(1,−1,1), so dot with c gives 1+k.

Worked derivation:

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} = (1, -1, 1) \cdot (1, 0, k) = 1 + k = 0 \Rightarrow k = -1.$$

Why: The scalar triple product is the signed volume of the vectors' parallelepiped. Vanishing volume means linear dependence and thus a common plane through the origin.

Why not the others: (A) reverses the needed sign; (B) leaves volume one; (C) yields negative volume.

Key point: Set the triple product to zero rather than its magnitude to one.

Q44 / A mean deviation of grouped observations

Answer: (A) 7/8.

Given: Four observations counted with multiplicity.

Formula/Principle: Mean=Σfx/Σf; mean deviation=Σf|x−mean|/Σf.

Step 1: Mean=(1+4+4)/4=9/4.

Worked derivation:

$$\text{MD} = \frac{|1 - 9/4| + 2|2 - 9/4| + |4 - 9/4|}{4} = \frac{5/4 + 2/4 + 7/4}{4} = \frac{7}{8}.$$

Why: Frequencies weight both the mean and absolute deviations. Squaring instead of taking absolute values would calculate a variance-like quantity rather than the statistic requested.

Why not the others: (B) ignores one deviation; (C) uses an unweighted average; (D) reports the mean.

Key point: Use the same frequency weights at both steps.

Q45 / B inverse-value branch

Answer: (B) $5\pi/6$.

Given: The original angle $7\pi/6$ is in quadrant III, but the three inverse functions have different principal ranges.

Formula/Principle: Reduce each function value into its own inverse range before adding.

Step 1:

$$\arcsin(\sin 7\pi/6) = \arcsin(-1/2) = -\pi/6.$$

Step 2:

$$\arccos(\cos 7\pi/6) = \arccos(-\sqrt{3}/2) = 5\pi/6.$$

Step 3:

$$\arctan(\tan 7\pi/6) = \arctan(1/\sqrt{3}) = \pi/6.$$

Step 4:

$$-\pi/6 + 5\pi/6 + \pi/6 = 5\pi/6.$$

Worked derivation:

$$\text{sum} = 5\pi/6.$$

Why: An inverse trigonometric function returns the principal representative, not an unreduced original angle. A common periodic adjustment cannot be used for all three terms because their inverse ranges differ.

Why not the others: (A) loses a term; (C) retains the original third-quadrant angle; (D) uses an incorrect tangent representative.

Key point: Apply each inverse range independently.

Physics

Q46 / C normal force under noninertial motion

Answer: (C) 24 N.

Given: Lift and block accelerate upward together.

Formula/Principle: N−mg=ma.

Step 1: N=m(g+a)=2(10+2)=24 N.

Worked derivation:

$$N - mg = ma \quad \Rightarrow \quad N = 2(10+2) = 24\,\mathrm{N}.$$

Why: Upward acceleration requires an upward net force, so the floor's normal must exceed weight. The block is stationary relative to the cabin but accelerated in the ground frame.

Why not the others: (A) assumes downward acceleration; (B) uses weight alone; (D) adds the acceleration twice.

Key point: Choose a frame and apply Newton's second law consistently.

Q47 / D energy with dissipative stage

Answer: (D) $50x^2 + 2x = 14$, with $x \approx 0.510$ m.

Given: $m = 1$ kg, $h = 2$ m, $\mu = 0.20$, $k = 100\ \mathrm{N\,m^{-1}}$, $g = 10\ \mathrm{m\,s^{-2}}$.

Formula/Principle: Initial gravitational energy equals spring energy plus friction work over the entire rough distance.

Step 1:

$$mgh = 20\ \mathrm{J}, \quad W_{\text{approach}} = \mu mg(3) = 6\ \mathrm{J}.$$

Step 2: Compression adds $W_{\text{compression}} = \mu mgx = 2x$ J and spring energy $kx^2/2 = 50x^2$ J.

Step 3:

$$20 = 6 + 2x + 50x^2 \Rightarrow 50x^2 + 2x - 14 = 0.$$

Step 4:

$$x = \frac{-2 + \sqrt{2^2 + 4(50)(14)}}{100} = \frac{\sqrt{701} - 1}{50} \approx 0.510\ \mathrm{m}.$$

Worked derivation:

$x > 0$ selects the positive quadratic root.

Why: The spring is on the same rough surface, so friction continues to remove energy while compression increases. The negative root is algebraically possible but cannot represent the stated nonnegative compression.

Why not the others: (A) omits all friction; (B) omits friction during compression; (C) confuses the fixed approach loss with the coefficient of variable compression distance.

Key point: Include dissipative work over each rough segment before solving for a turning point.

Q48 / A gravity inside uniform sphere

Answer: (A) One third.

Given: Spherical uniform density; interior enclosed mass scales as r^3.

Formula/Principle: $g(r)=GM(r)/r^2$, with $M(r)=M(r/R)^3$.

Step 1: $g(r)/g(R)=r/R$; at $r=R/3$ the ratio is 1/3.

Worked derivation:

$$\frac{g(R/3)}{g(R)} = \frac{M(R/3)/(R/3)^2}{M/R^2} = \frac{(1/3)^3}{(1/3)^2} = \frac{1}{3}.$$

Why: Using the entire planet mass at an interior point would be wrong: outer spherical shells contribute no net field there. Only the enclosed mass shrinks with radius, leaving a linear dependence.

Why not the others: (B) uses inverse-square distance with fixed mass; (C) assumes surface field throughout; (D) reverses the linear ratio.

Key point: Apply the shell theorem before Newton's inverse-square law.

Q49 / B stress-strain two regimes

Answer: (B) 50 GPa.

Given: Elastic endpoint stress 100×10^6 Pa, strain 0.002.

Formula/Principle: Young modulus is slope $d\sigma/d\varepsilon$ in the elastic segment.

Step 1: $100\times10^6/0.002=50\times10^9$ Pa.

Worked derivation:

$$Y = \frac{\Delta\sigma}{\Delta\varepsilon} = \frac{100\,\text{MPa}}{0.002} = 50000\,\text{MPa} = 50\,\text{GPa}.$$

Why: A later plastic slope does not represent Young modulus of the initial reversible response. Strain is dimensionless, so the ratio has pressure units.

Why not the others: (A) divides by an incorrect strain; (C) uses the numerical stress as modulus; (D) applies a factor four.

Key point: Read slope only from the elastic part of the graph.

Q50 / C calorimetry with phase plateau

Answer: (C) About 63 g melts at 0°C.

Given: Water can release $100\times4.2\times50=21000$ J cooling to zero; melting all ice needs 33400 J.

Formula/Principle: At phase plateau, available heat melts mass $m=Q/L$.

Step 1: $21000/334\approx62.9$ g, leaving about 37.1 g ice.

Worked derivation:

$$m_{\text{melted}} = \frac{(100)(4.2)(50)}{334} \approx 62.9\,\text{g} < 100\,\text{g}.$$

Why: The warm water runs out of sensible heat before all ice melts. Coexisting ice and water fix final equilibrium at 0°C under the stated pressure assumption.

Why not the others: (A) and (B) require more melting heat than available; (D) ignores the positive heat transfer.

Key point: Test whether the phase change completes before solving for final temperature.

Q51 / D rms speed for gas mixtures

Answer: (D) √(5/2).

Given: Equal molecule counts, common temperature, and masses MX:MY=4:16.

Formula/Principle: Species mean-square speeds scale as 1/M; the pooled mean square is the count-weighted average.

Step 1: ⟨vX²⟩/⟨vY²⟩=MY/MX=4.

Step 2: ⟨v²⟩mix/⟨vY²⟩=(4+1)/2=5/2.

Worked derivation:

$$\frac{v_{\text{rms,mix}}}{v_Y} = \sqrt{\frac{\langle v_X^2\rangle + \langle v_Y^2\rangle}{2\langle v_Y^2\rangle}} = \sqrt{\frac{4+1}{2}} = \sqrt{\frac{5}{2}}.$$

Why: rms means square first, average across all molecules, then take the square root. Averaging the two rms speeds directly would be mathematically different; equal mole fractions justify equal weights.

Why not the others: (A) reverses the mass trend; (B) discards X's larger speeds; (C) uses an incorrect weighted square.

Key point: Combine mean-square speeds before taking the mixture rms.

Q52 / A Infer lift acceleration from period before calculating support force

Answer: (A) $a/g = 1$; scale reading $2mg$.

Given: Q experiences effective downward acceleration $g + a$; its length is $L/2$.

Formula/Principle:

$$T = 2\pi\sqrt{\ell/g_{\text{eff}}}, \qquad N - mg = ma.$$

Step 1:

$$\left(\frac{T_Q}{T_P}\right)^2 = \frac{(L/2)/(g+a)}{L/g} = \frac{g}{2(g+a)} = \frac{1}{4}.$$

Step 2:

$$4g = 2(g+a) \Rightarrow a = g.$$

Step 3:

$$N = m(g+a) = 2mg.$$

Worked derivation:

$$a/g = 1, \qquad N = 2mg.$$

Why: The observed period ratio determines the effective acceleration only after the length ratio is included. The same inferred lift acceleration then enters an independent force balance for the scale reading.

Why not the others: (B) gives a period ratio $1/\sqrt{3}$; (C) gives $1/\sqrt{2}$; (D) gives $1/\sqrt{8}$, none matching the observed one half.

Key point: Infer an accelerated frame's gravity from oscillation data before predicting support force.

Q53 / B series-capacitor charge

Answer: (B) 3/2.

Given: Initially C and 2C in series; after insertion they are 2C and 2C in series with fixed battery V.

Formula/Principle: Q=CeqV for each state.

Step 1: Ceq,old=2C/3; Ceq,new=C.

Worked derivation:

$$\frac{Q_{\text{new}}}{Q_{\text{old}}} = \frac{CV}{(2C/3)V} = \frac{3}{2}.$$

Why: The connected battery supplies additional free charge as total capacitance increases. Treating the original series charge as fixed would apply a disconnected-circuit condition instead.

Why not the others: (A) ignores dielectric insertion; (C) reverses the ratio; (D) uses capacitance of one plate pair as the initial equivalent.

Key point: A connected ideal battery fixes voltage across the series combination.

Q54 / C magnetic dipole energy

Answer: (C) +mB.

Given: θ changes from 0 to π/2.

Formula/Principle: U=−mB cosθ.

Step 1: Uinitial=−mB, Ufinal=0; ΔU=+mB.

Worked derivation:

$$\Delta U = -mB\cos(\pi/2) - [-mB\cos 0] = 0 + mB = mB.$$

Why: Parallel orientation is lowest energy, and rotating against the aligning tendency raises potential energy. The sign belongs to final minus initial, not the torque direction.

Why not the others: (A) reverses endpoints; (B) omits orientation dependence; (D) uses an antiparallel final state.

Key point: Evaluate both orientation energies before subtracting.

Q55 / D dipole torque and energy

Answer: (D) Torque zero, energy lower.

Why: Torque magnitude is pE sinθ, largest at 90° and zero at alignment. Potential U=−pE cosθ falls from zero to −pE, with the released energy available for rotation.

Why not the others: (A) reverses both trends; (B) retains nonzero aligned torque; (C) gives the wrong energy sign.

Worked derivation:

$$\tau_i = pE,\ \tau_f = 0; \qquad U_i = 0,\ U_f = -pE.$$

Key point: Torque and potential have different angular functions.

Q56 / A bridge balance with temperature

Answer: (A) 50°C.

Given: Balance ratio 2/3=4/R; R(T)=5(1+0.004T).

Formula/Principle: Solve bridge balance, then temperature-resistance relation.

Step 1: R=6 Ω; 6/5−1=0.2=0.004T.

Worked derivation:

$$\frac{2}{3} = \frac{4}{R} \Rightarrow R = 6\,\Omega; \quad 6 = 5(1 + 0.004T) \Rightarrow T = 50^{\circ}\text{C}.$$

Why: At balance the bridge detector current vanishes, setting arm ratios. Heating raises the lower-right resistance from five to six ohms; solving the material law afterwards avoids mixing circuit and

temperature effects.

Why not the others: (B) yields 5.5 Ω; (C) yields 6.5 Ω; (D) yields 7 Ω.

Key point: Solve the electrical target resistance first.

Q57 / B critical angle through layers

Answer: (B) Grazing emergence.

Given: Parallel interfaces; nA sin30°=2×1/2=1.

Formula/Principle: Snell's law keeps n sinθ constant through parallel layers.

Step 1: In B, sinθB=1/√2, θB=45°; in air, sinθair=1, θair=90°.

Worked derivation:

$$2\sin 30^\circ = \sqrt{2}\sin 45^\circ = 1\sin 90^\circ = 1.$$

Why: The ray reaches the critical angle at the last interface, so transmitted direction lies along the boundary in the ideal geometric limit. It is not beyond critical, so ordinary total internal reflection is not the correct description.

Why not the others: (A) ignores transmission into B; (C) carries the original angle without Snell correction; (D) copies the B angle into air.

Key point: Propagate the conserved n sinθ across both boundaries.

Q58 / C stopping potential from two frequencies

Answer: (C) 2 eV.

Given: Graph slope (1 V)/(2.5×10¹⁴ Hz)=4×10⁻¹⁵ V s, equal to h/e.

Formula/Principle: Threshold $hf_0=\varphi$; in eV, $\varphi/e=(h/e)f_0$.

Step 1: (4×10⁻¹⁵)(5×10¹⁴)=2 V numerically, hence 2 eV.

Worked derivation:

$$\frac{\phi}{e} = \frac{h}{e}f_0 = \frac{1\,\mathrm{V}}{(7.5-5)\times 10^{14}\,\mathrm{Hz}}(5\times 10^{14}\,\mathrm{Hz}) = 2\,\mathrm{V}.$$

Why: The graph's zero intercept supplies threshold frequency and its slope supplies Planck's constant per electron charge. Both are needed; a stopping potential of one volt at one frequency is not the work function.

Why not the others: (A) uses a wrong frequency ratio; (B) copies the plotted voltage; (D) doubles the slope result.

Key point: Read slope and threshold separately from a photoelectric graph.

Q59 / D Path mixture inferred from photon count and mean photon energy

Answer: (D) Direct fraction one half; mean photon-energy ratio two thirds.

Given: A direct path emits one photon, a cascade emits two; each atom has the same initial and final energy levels.

Formula/Principle: Count paths from total photon number, then use conservation of energy across each complete path.

Step 1: If n atoms take the direct path,

$$n + 2(100-n) = 150 \Rightarrow n = 50.$$

Step 2: Every path releases $E_A = E_4 - E_2$, because $(E_4 - E_3) + (E_3 - E_2) = E_4 - E_2$.

Step 3: Total photon energy is $100E_A$, shared by 150 photons, so

$$\overline{E}/E_A = 100/150 = 2/3.$$

Worked derivation:

$$f_{\text{direct}} = 50/100 = 1/2, \qquad \overline{E} = \frac{2}{3}E_A.$$

Why: Equal total energy per atom does not imply equal energy per emitted photon when the paths have different photon counts. The count observation identifies the path mixture before averaging energy over photons rather than atoms.

Why not the others: (A) averages over atoms rather than photons; (B) gives the wrong path count; (C) reverses the photon-to-energy averaging ratio.

Key point: Identify the counting unit in a mixed radiative-path average.

Q60 / A rotating rod motional emf

Answer: (A) 1.0 V.

Given: Rod length 1 m, angular speed 10 rad s^{-1}, field 0.20 T.

Formula/Principle: Element at radius r has speed ωr and incremental emf dε=Bωr dr.

Step 1: Integrate from pivot 0 to L: ε=½BωL².

Worked derivation:

$$\varepsilon = \int_0^L B\omega r\, dr = \frac{1}{2}(0.20)(10)(1)^2 = 1.0\,\text{V}.$$

Why: Speed increases along the rod, so replacing all elements by tip speed overestimates emf by a factor two. The perpendicular field makes the radial motional-force component maximal.

Why not the others: (B) uses tip speed everywhere; (C) omits angular speed; (D) omits both the half and B.

Key point: Integrate local motional emf over a rotating rod.
